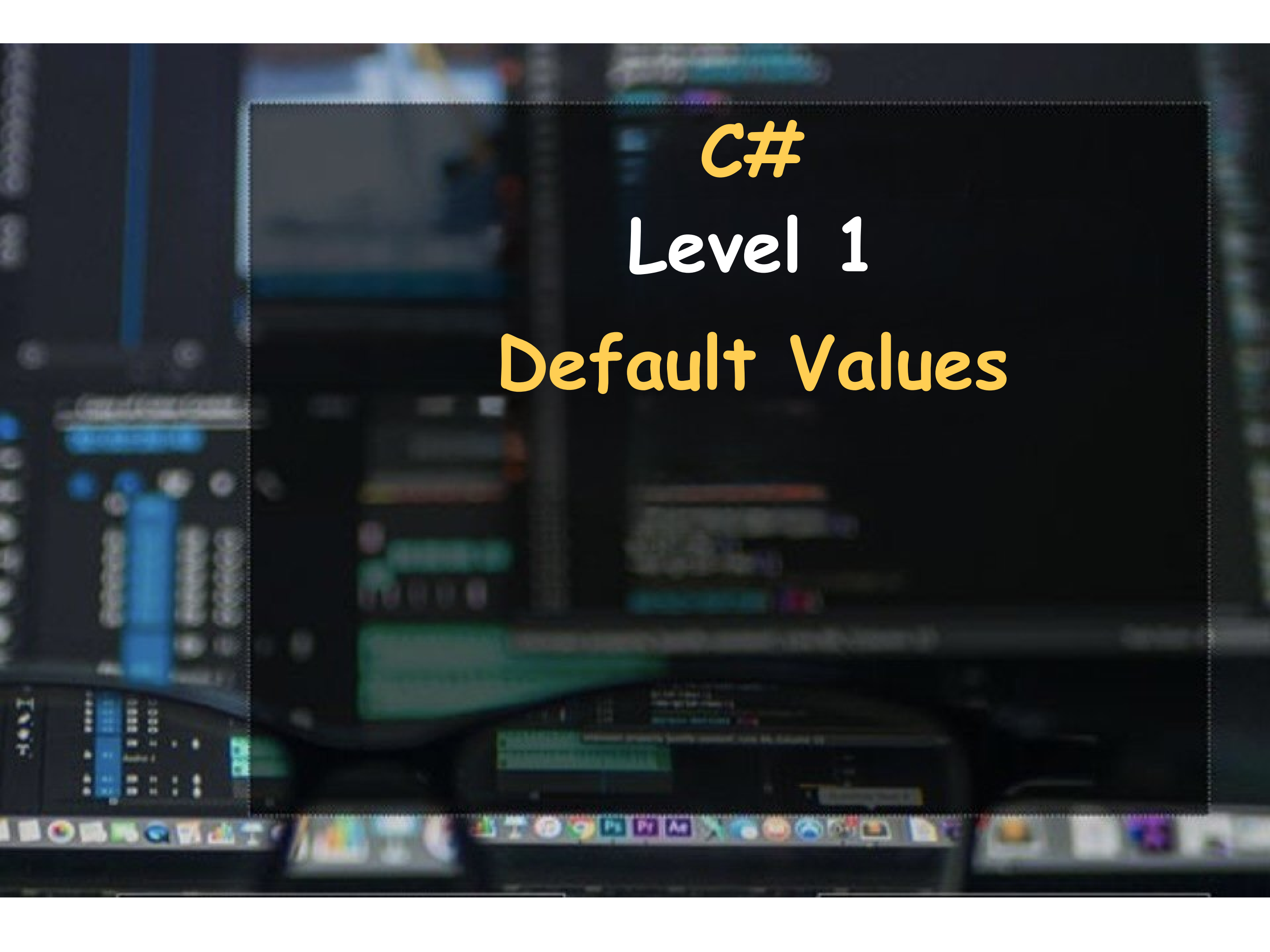
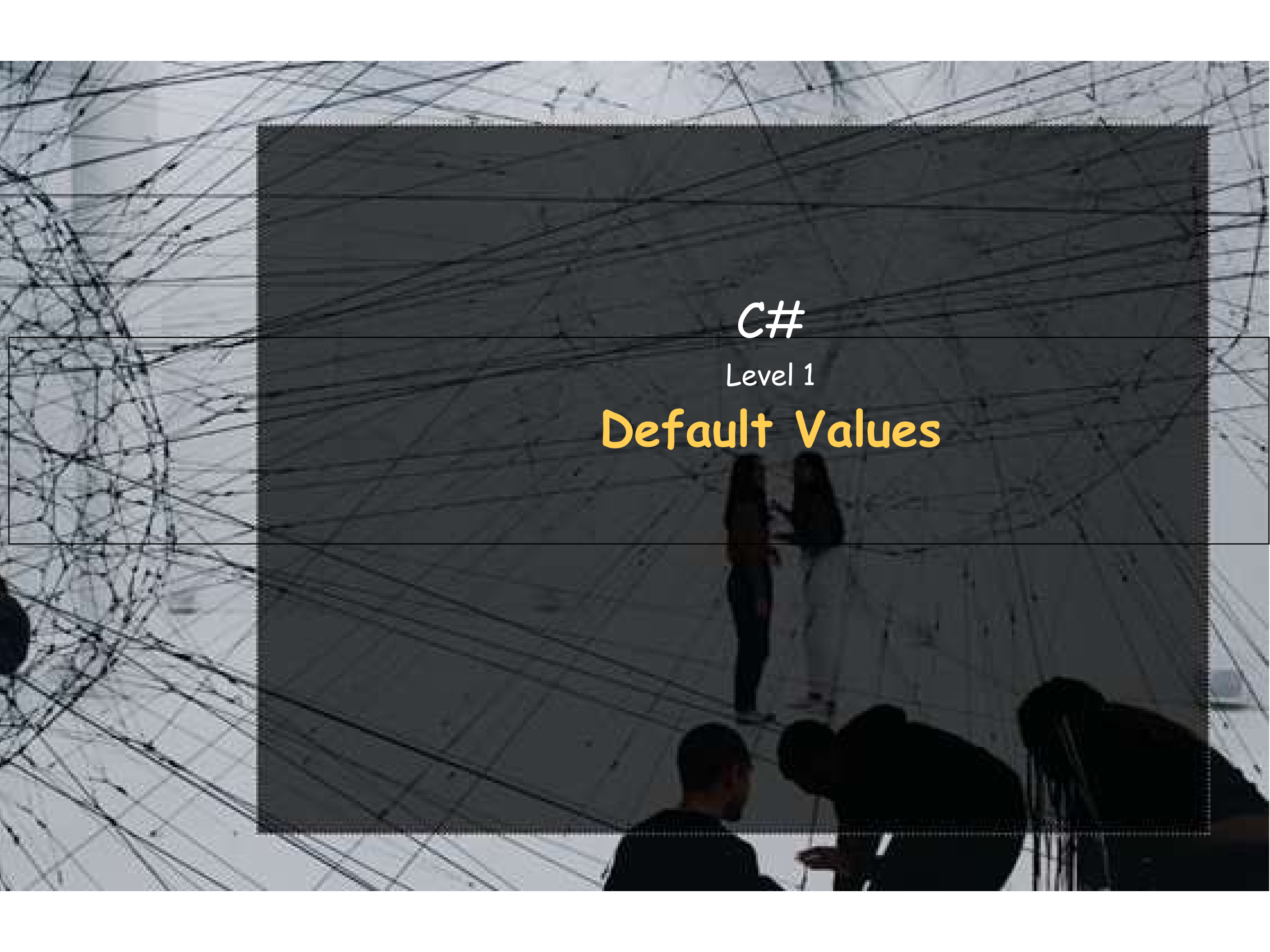




# C#

## Level 1

# Default Values



C#

Level 1

# Default Values

# Definition

## Default Value

-  `default` returns the default value of a specified data type.

-  The default value depends on the type of the variable.

## Explanation

-  Using ``default(type)``

-  The type can be specified explicitly inside ``default()``.

# Procedures

1. Declare a variable with its data type.
2. Assign ``default(type)`` or ``default``.
3. C# assigns the default value for that type.
4. Use the variable normally.

# 💡 Example



```
int i = default(int);  
float f = default(float);  
decimal d = default(decimal);  
bool b = default(bool);  
char c = default(char);
```

➡ Result

```
int    → 0  
float  → 0  
decimal → 0  
bool   → false  
char   → '\0'
```



```
int i = default;  
float f = default;  
bool b = default;
```

➡ Result

```
int    → 0  
float  → 0  
bool   → false
```

## Note

◆ Common default values:

▣ Numeric types → `0`

▣ `bool` → `false`

▣ `char` → `'\0'`

▣ Reference types → `null`

◆ `default` does not mean a random or empty value; it returns the predefined default for that type.

## ⊘ Warning

- ◆ `default` without a type requires the compiler to know the variable's type.  
    `int number = default;`
- ◆ `default(type)` explicitly specifies the type:  
    `int number = default(int);`

# Comparison

## ``default(type)``

- ✓ Type is explicitly specified.
- ✓ Example: ``default(int)``.

## ``default``

- ✓ Type is inferred from the context.
- ✓ Introduced for this syntax in *C# 7.1*.
- ✓ Example: ``int number = default;``.

# ✦ Summary

- ✓ ``default`` returns the default value of a type.
- ✓ Numeric types → ``0``.
- ✓ ``bool`` → ``false``.
- ✓ ``char`` → ``'\0'``.
- ✓ Reference types → ``null``.
- ✓ ``default(type)`` specifies the type explicitly.
- ✓ ``default`` lets the compiler determine the type from context.

Thank you 😊